

## M003 Extension 2 — English Worksheet

**Topic:** 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Extension (after Week 3) · **Time:** about 45 minutes

Build **two mob heads** in 3D — a **creeper** and a **piglin**. Now every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **forward** (depth). Both heads stand in front of you in the world — walk around them, then copy one block by block.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

### 1 · Predict 🧐

Read each set. Write what you will see.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. Does the line go across, up, or deeper?

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```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. Across, up, or deeper?

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```
place green block at (3, 5, 1)
place black block at (3, 5, 1)
```

**Same (x, y, z). Green, black, or both? Why?**

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## 2 · Spot the Bug 🐛

Fix each one. Then say why it was wrong.

**Bug A** — This block should sit one step **deeper**, at (3, 5, 2). It only has two numbers, so it lands on the front.

```
place green block at (3, 5)
```

**Fixed code:**

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**Why wrong? Why does your fix work?**

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**Bug B** — The eye should be **up high and one step deep**: (4, 7, 2). The last two numbers are swapped.

```
place black block at (4, 2, 7)
```

**Fixed code:**

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**Why wrong? Why does your fix work?**

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**Bug C** — Code is correct, but the whole head came out one step too far forward.

(the code matched the head)

**What did your friend forget before pressing run?**

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**Start with the front face** (keep  $z$  the same, e.g.  $z = 1$ ). Read each block's **x** (across) and **y** (up).

💡 Tip: the eyes and mouth only show on the **front**. The sides and back are plain.

I built the ...

The lowest front block is at (x = ..., y = ..., z = ...).

To go up I changed the ... number; to go deeper I changed the ... number.

My head is ... blocks deep.

Once my head was wrong because ...

Next time I would ...

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## 4 · Record Your Walkthrough 🎥

Film yourself showing your head. Use these words: **x, y, z, across, up, forward, home spot.**

1. Show your home spot. Why stand on it every time?
2. Walk around your head — show the front, a side, and the back.
3. Point at one block: "this one is at x ..., y ..., z ...".
4. Show one block you fixed.

**Plan what you will say:**

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